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Project Title:	The CQ Experiment		
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Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Low temperature condensed matter physics		
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Zip Code:	91125	**Congressional District:**	29
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The heat capacity of superfluid helium in a constant heat flux has been predicted to become infinite at a critical value of the heat flux where superflow becomes unstable. Measurements done in our laboratory at Caltech have succeeded in detecting an excess heat capacity due to the presence of a heat flux but, for a variety of gravity-related reasons, it is impossible in the laboratory to make measurements at values of the heat flux that approach the critical value. The CQ Experiment will measure the heat capacity (C) at constant heat flux (Q) in the absence of gravity. The CQ Experiment will be a guest experiment, using the hardware of Critical Dynamics in Microgravity (DYNAMX) without engineering modification. The DYNAMX Instrument Sensor Package (ISP) was not designed for precision calorimetry, but is nearly ideal for that purpose. Scheduled for the M1 mission aboard the Low Temperature Microgravity Physics Facility (LTMPF), DYNAMX should require about one month to complete its measurements. The liquid helium on the M1 mission is expected to last 4-5 months. The CQ experiment will be done in the DYNAMX ISP during the remaining time.
The CQ Experiment will measure the heat capacity of superfluid helium at constant heat flux, over a range of temperatures within one millionth of a Kelvin of the superfluid transition (the lambda point, where helium becomes superfluid in the absence of a heat flux). The range of heat fluxes will be from zero up to a few millionths of a Watt per square centimeter. Only in these conditions can helium conduct heat at constant temperature (within the 50 pico-Kelvin resolution of our thermometry). At higher heat fluxes, temperature gradients would render heat capacity measurements meaningless. With these small heat fluxes the experimental signal, excess heat capacity due to the heat flux, becomes resolvable only within one micro-Kelvin of the lambda point.

Task Description:

The goal of the experiment is to measure the heat capacity as close as possible to the critical value of the heat flux, where superflow becomes unstable, and where the heat capacity is predicted to approach infinity.

On Earth, the hydrostatic pressure of a column of liquid helium causes the lambda point temperature itself to vary by more than one micro-Kelvin per centimeter of column height. To combat this problem, our laboratory measurements were made in a cell only 0.06 centimeters high. Even so, all our data are severely compromised by the hydrostatic gravity effect. In addition, the very small cell height made it necessary to rely on temperatures measured in the end caps of the cell, greatly complicating data analysis and limiting the range of measurements to values of the heat flux far below the critical value. All or most of these problems will be overcome by making the measurements in the DYNAMX ISP. The DYNAMX cell is 11 mm high and measures the temperature with high resolution thermometry at three positions along the side wall. There are two outstanding discrepancies in this field at the moment. The first is that experiment and theory disagree on the value of the critical heat current as a function of temperature, and the other is that the excess heat capacity measured in our own experiments is much larger than that predicted theoretically. A successful CQ experiment would shed considerable light on the causes of both of these discrepancies. If in addition it turns out to be possible to measure the heat capacity close enough to the critical heat current to observe the functional form of the predicted divergence, that result might very well open a whole new field of physics. The form of the divergence should be dominated by fluctuations of a qualitatively different kind from those that govern the behavior of conventional critical points.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Regrettably, this last year has seen the cancellation of the DYNAMX/CQ flight projects due to NASA's new Moon and Mars initiative. This has necessitated a consolidation of the resources of our two teams. To this end some of the DYNAMX flight hardware testing and the ongoing DYNAMX/CQ ground science has moved from the University of New Mexico (UNM) to Caltech. This also coincides with Prof. Robert Duncan (PI of DYNAMX) being appointed as a Gordon and Betty Moore Distinguished Scholar at Caltech (from August 2004 to 2005).
We now have two experiments at Caltech focusing on different aspects of the CQ science. The first experiment is designed to study the Self Organized Critical (SOC) state, an exotic state of matter that exists throughout nature. In this state the system, though subject to external forces that would normally drive it away from equilibrium, manages to exist in a steady-state close to its phase transition, or "critical line". In our case the system is liquid helium, the critical line is the superfluid transition (T-lambda) and the external force is the heat flux (Q) applied to the helium sample. As the helium enters this state each point in the sample 'self-organizes' to have the same temperature difference from T-lambda.

We have recently succeeded in making the first measurements of the heat capacity of the fully developed SOC state. The preliminary results of this work will be presented in August 2005 at the 24th International Conference on Low Temperature Physics in Orlando, Florida, USA. It is hoped that these measurements will elucidate the physics of the CQ experiment as it approaches its critical point.

Task Progress:

Our second experiment is still being set up within a pre-existing cryoprobe that was transported from the UNM to Caltech in September, 2004. This apparatus contains all of the thermal isolation and control technology that has been designed for the flight experiment, including, at its heart, the experimental cell that has been constructed to flight specifications. In order to ameliorate the effect of gravity, this cell has been made shorter than the planned flight cell.

This is the first fully 'flight-like' cell to be built by DYNAMX and required over two work-years of effort to complete. It incorporates improvements in low temperature thermometry and construction techniques, some of which were a direct consequence of the DYNAMX/CQ project. One example of this is a novel design of one endplate that serves to spread the heat flux evenly before it enters the helium sample. In past experiments, inhomogeneous heat flow may have caused erroneous measurements of the heat capacity. This work will be presented at the 24th International Conference on Low Temperature Physics.

With this apparatus the DYNAMX/CQ teams hope to make new measurements of CQ and related phenomena, very near to the superfluid transition. From these new, high fidelity, measurements we hope to reconfirm our previous ground-based measurements, including those of other authors, and clarify some of the discrepancies between those previous measurements and theory. Once this is accomplished these measurements may very well represent just about the best that can be achieved on Earth under the deleterious effect of gravity.

In addition: Recent theoretical work, conducted primarily by Prof. Stephen Boyd (UNM, DYNAMX team member) may help in understanding the observation, by R. A. M. Lee et al., JLTP, 134, Nos. 1/2, 495-505 (2004), of a heat flow asymmetry in the thermal resistance of superfluid helium near its transition temperature. This work was presented by

	Prof. Boyd at Caltech, in 2004 and at the March Meeting of the American Physical Society, in 2005. It has also been accepted as a contributed oral presentation at the 24th International Conference on Low Temperature Physics.
Bibliography Type:	Description: (Last Updated: 02/25/2014)
Papers from Meeting Proceedings	Liu J, Sergatskov DA, Duncan RV. "Adaptive optimal PI controller for high-precision low-temperature experiments." American Control Conference 2005, Portland, Oregon, June 8-10, 2005. Proceedings of the American Control Conference 2005, Portland, Oregon, June 8-10, 2005. volume 6, p. 4220- 4224. http://dx.doi.org/10.1109/ACC.2005.1470641 , Jun-2005
Presentation	Duncan, R. V. "Ultra-High-Resolution Studies of (at least one) Non-Equilibrium Quantum Phase Transition" Physics Research Conference, California Institute of Technology. Oct-2004
Presentation	Boyd, S. T. P. "Temperature Profiles of Hydrodynamic Origin in Counterflowing Helium-II near T-lambda" Condensed matter physics seminar, California Institute of Technology. Nov-2004
Presentation	Boyd, S. T. P. "Temperature Profiles of Hydrodynamic Origin in Counterflowing Helium-II near T-lambda" March Meeting of the American Physical Society. Mar-2005